

2	B	The object starts from rest and moves along a straight line. Area under the acceleration-time graph represents <u>change</u> in velocity. The triangle below the time axis shows that the change in velocity is in the negative direction. Since the <u>initial velocity is zero</u>, that triangle represents the increasing velocity (in – ve direction). This happens up to the time at A. Beyond A, the acceleration changes direction (from – ve to + ve dir). However, the velocity still points in the – ve direction after A i.e. object continues to travel in the same direction (still getting further away from the starting point), except that since the acceleration and velocity point in opposite directions, the object is slowing down. At point B the object comes to a momentary stop and it is also the furthest from the starting point. Between B and C, the object changes its velocity direction and starts to speed up back towards the starting point. At C, the object possesses maximum speed as it moves towards the starting point. Beyond C, the object slows down but it is still moving towards the starting point. At D, the object is back at the starting point.
3	D	$s = s_0 + v_0 t + \frac{1}{2} a t^2$